

Movie Success Prediction & Sentiment Study

Project 8 | Domain: Data Analytics & Machine Learning | Tools: Python, NLTK, VADER, Scikit-learn, XGBoost, Excel

1. Introduction

The film industry generates billions of dollars annually, yet predicting a movie's commercial success and audience reception remains a significant challenge. This project addresses that challenge by combining financial analytics with Natural Language Processing (NLP) to predict box office revenue and audience ratings using historical movie data. By leveraging the TMDB 930K+ Movies Dataset sourced via Kaggle Hub, the project builds a comprehensive intelligence system that forecasts performance metrics and performs sentiment analysis on plot overviews and audience reviews.

2. Abstract

This project implements a dual-track machine learning pipeline to predict (i) box office revenue and (ii) audience vote ratings for movies. The dataset comprises 930,000+ records filtered to movies with budgets above \$50,000 and non-zero revenue figures, yielding a high-quality analytical sample. Feature engineering included financial metrics (ROI, net profit, profit margin), TF-IDF vectorization of plot overviews (250 features, 1–2 ngrams), genre one-hot encoding, and VADER-based sentiment scoring. XGBoost Regressor was deployed for revenue forecasting while a Random Forest Ensemble targeted rating prediction. Additionally, a content-based movie recommendation system was constructed using cosine similarity on a metadata-soup combining genres, production companies, and keywords. Results were compiled into an Excel analytical portfolio with model diagnostics, genre-wise sentiment trends, and a live prediction sandbox.

3. Tools Used

- **Python 3.x** — Core programming language for data processing, modelling, and visualization
- **Pandas & NumPy** — Data loading, cleaning, type enforcement, and feature engineering
- **NLTK & VADER** — Tokenization, stopword removal, lemmatization, and sentiment scoring
- **Scikit-learn** — TF-IDF vectorization, train-test split, StandardScaler, RandomForestRegressor
- **XGBoost** — Gradient-boosted regressor for box office revenue prediction
- **Matplotlib & Seaborn** — EDA visualizations — scatter plots, bar charts, heatmaps
- **KaggleHub API** — Programmatic download of the TMDB 930K+ movies dataset
- **Excel (openpyxl)** — Final analytical report with multiple sheets for model diagnostics and trends

4. Steps Involved in Building the Project

Step 1. Data Acquisition

Downloaded the TMDB 930K+ Movies Dataset via Kaggle Hub API. Filtered records to movies with budget > \$50,000 and revenue > \$0 to eliminate noise.

Step 2. Data Cleaning & Type Enforcement

Applied robust type casting to numeric columns (budget, revenue, popularity, vote_average, runtime). Filled missing runtime values with the column median. Created defensive copies to avoid chain-assignment issues.

Step 3. Feature Engineering

Derived financial features: net_profit, ROI, profit_margin, budget_tier. Parsed JSON-encoded genre and production company fields. Extracted release year and month from date strings.

Step 4. Sentiment Analysis

Used NLTK's VADER SentimentIntensityAnalyzer on movie plot overviews. Generated compound sentiment scores and averaged them per movie for an audience_sentiment feature.

Step 5. TF-IDF Vectorization

Applied TF-IDF on plot overviews with 250 max features and 1–2 ngrams to capture key themes. Merged the sparse TF-IDF matrix with the main feature dataframe.

Step 6. ML Track 1 — Revenue Prediction

Applied log1p transformation on revenue to handle skew. One-hot encoded genres. Trained XGBoost Regressor with 80/20 split. Evaluated using R² and RMSE.

Step 7. ML Track 2 — Rating Prediction

Isolated vote_average as the target. Trained Random Forest Regressor on scaled features. Reported R² and RMSE metrics.

Step 8. Recommendation System

Built a content-based recommender using cosine similarity on a metadata soup (genres, production companies, keywords). Top-5 similar movies returned per query.

Step 9. Live Prediction Sandbox

Implemented an interactive module accepting custom movie parameters (title, budget, popularity, runtime, sentiment, genre) to output predicted revenue and rating in real time.

Step 10. Excel Report Generation

Exported results to a multi-sheet Excel workbook containing: ML model diagnostics, genre-wise sentiment trends, top-performing movies, and recommendation samples.

5. Conclusion

The Movie Success Prediction and Sentiment Study successfully demonstrated how financial data and NLP-derived features can jointly forecast movie performance. The XGBoost model achieved strong R² scores on revenue prediction (log-space), while the Random Forest model effectively captured rating variance. Genre-wise sentiment analysis revealed that Drama and Documentary films scored highest in positive audience sentiment, while Horror and Thriller genres showed more polarized responses. The content-based recommendation system provided accurate suggestions based on thematic and production profile similarities. This project reinforced key data analytics competencies including feature engineering, model evaluation, NLP pipeline construction, and result communication through structured Excel reports — all directly applicable to professional Data Analyst and ML Engineer roles.